# LINEAR ALGEBRA AND ITS APPLICATIONS

Department of Science and Humanities

# SPECTRAL THEORM

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## SPECTRAL THEORM -STATEMENT

An $n \times n$ symmetric matrix A has the following properties:

a. A has n real eigenvalues, counting multiplicities.

b. The dimension of the eigenspace for each eigenvalue λ equals the multiplicity of λ as a root of the characteristic equation.

c. The eigenspaces are mutually orthogonal, in the sense that the eigenvectors corresponding to Different eigenvalues are orthogonal.

d. A is orthogonally diagonalizable.

# SPECTRAL DECOMPOSITION THEORM -STATEMENT

Let A be a $n \times n$ matrix. Let $\{u_1, u_2, \dots, u_n\}$ be an orthonormal basis for $R^n$ consisting of eigenvectors of A with corresponding eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$.Then there exists Symmetric matrices $P_1, P_2, \dots, P_n$ such that the following results hold:

*i)* $A = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_n P_n$.

ii) Rank $P_i = 1 \; for \; all \; i$.

*iii)* $P_i \, P_i = P_i \;\; for \; all \; i \;, and \; P_i \, P_j = 0 \; for \; i \neq j$.

*iv)* $P_i \, u_i = u_i \;\; for \; all \; i$ , and $P_i \, u_j = 0 \; for \; all \; i \neq j$.

# SPECTRAL THEORM -EXAMPLE

Let $A = \begin{pmatrix} 3 & -4 \\ -4 & 3 \end{pmatrix}$

The eigenvalues of A are $\lambda_1 = 5, and\ \lambda_2 = -5.$

The orthonormal eigenvectors are $u_1 = \frac{1}{\sqrt{5}}\begin{pmatrix} -2 \\ 1 \end{pmatrix}, u_2 = \frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ 2 \end{pmatrix}.$

$$P_1 = u_1 u_1^T = \begin{bmatrix} 4/5 & -2/5 \\ -2/5 & 1/5 \end{bmatrix}$$

$$P_2 = u_2 u_2^T = \begin{bmatrix} 1/5 & 2/5 \\ 2/5 & 4/5 \end{bmatrix}$$

We can verify that

$$A = \lambda_1 P_1 + \lambda_2 P_2$$

# Additional problem

- Find a spectral decomposition of the matrix $A = \begin{bmatrix} 4 & 1 & -1 \\ 1 & 4 & -1 \\ -1 & -1 & 4 \end{bmatrix}$.

Solution:

The eigenvalues are 3,3,6.

The orthonormal basis consisting of corresponding eigenvectors are

$$\left(\frac{1}{\sqrt{2}},0,\frac{1}{\sqrt{2}}\right),\left(-\frac{1}{\sqrt{6}},\frac{2}{\sqrt{6}},\frac{1}{\sqrt{6}}\right),\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},-\frac{1}{\sqrt{3}}\right)$$

The spectral decomposition of A is given by

$A = \lambda_1 P_1 + \lambda_2 P_2 + \lambda_3 P_3$, where $P_i = u_i u_i^T$

# THANK YOU

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